

Vincent S. Pronel

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EDUCATION

Bachelor of Science in Computer Science

Kean University

Expected 2027

Union, NJ

- **Relevant Coursework:** Database Management Systems, Applied Statistics, Matrix & Linear Algebra, Analysis of Algorithms*, Software Engineering*, Object Oriented Analysis & Design **in progress, Fall 2026*
- **Leadership & Honors:** Web Developer, WiCyS (Women in Cyber Security) Kean Chapter E-Board | GMIS Research Posters Competition | Presenter, 2026 CRA UR2PhD Research Showcase, New Orleans | Dean's List, Spring 2024 and Fall 2025

TECHNICAL SKILLS

- **Data & Databases:** SQL, MySQL, PostgreSQL, Supabase, Cloudflare D1, relational schema design, data marts, aggregation and normalization, correlation analysis
- **Visualization:** Google Charts, interactive dashboards, scatter plots and trendlines, correlation matrices, chart design
- **Languages:** SQL, PHP, Python, JavaScript, TypeScript, Java, HTML/CSS
- **Platforms & Tools:** MySQL Workbench, Apache/XAMPP, Next.js, React, Node.js, REST APIs, Cloudflare, Vercel, Git/GitHub, Linux (Ubuntu), Claude Code, Codex, Figma

RESEARCH PROJECTS

Java Execution Visualizer | CRA UR2PhD-REU Fellowship

June 2026 – Present

Undergraduate Research Fellow, Advisor: Dr. Yan Ma | Competitively selected, \$8,000 funded, 400 hours

Kean University

- Developed a tool that visualizes Java program state step by step, integrating a JDI-based tracer and rendering variables, objects, the call stack, and the heap at every executed line.
- Designed the **Supabase** schema and participant instrumentation capturing attempt and event data for an IRB-approved study measuring learning outcomes and cognitive load against static instructional materials.

Jobs vs. Wages | NSF IUSE Scholars Academy

May 2026 – Present

NSF IUSE Research Scholar, Advisor: Dr. Pauline Wang | Competitively selected, \$1,400 funded, 10 weeks

Kean University

- Extracted and normalized 33,000 ZIP-code records into population-weighted state-level **data marts** in **MySQL**, rolling county figures up to all 52 U.S. states and territories; designed the schema and the aggregation logic end to end.
- Developed an interactive dashboard in PHP, JavaScript, and **Google Charts** with selectable variable pairs, live scatter plots with linear trendlines and R^2 , a full correlation matrix, and a multi-state comparison view.
- Disproved the study's own premise: unemployment explains only 14% of variation in household income ($r = -0.38$), while college attainment ($r = 0.79$) and poverty rate ($r = -0.74$) are more than twice as predictive.

AI Debate Coach Evaluation | CRA UR2PhD Research Program

Feb. 2026 – Apr. 2026

Undergraduate Researcher, Advisor: Dr. Yan Ma

Kean University

- Designed and conducted the measurement framework for a 10-week study on LLM sycophancy, combining GPT-4-as-a-judge scoring with a 12-participant human study across three model configurations.
- Quantified a 100% source fabrication rate in the fine-tuned model that only 1 of 12 participants detected; humans preferred the prompted model 3 to 1 while the automated metric rated the two comparably, demonstrating that the automated evaluator would have selected the worse system.

PROFESSIONAL EXPERIENCE

Independent Web Developer

Jan. 2026 – Present

Freelance Full-Stack Web Development

Sayreville, NJ

- Founded and run a freelance practice serving 4 business clients across 4 live sites; design the **relational schemas** and data layer for every project, and build Supabase-backed admin editors that let non-technical owners maintain their own content.
- Live sites: daysicecream.com (4,000+ monthly visitors) | trainingaio.com (booking and payments schema, delivered in 2 weeks) | deafsharkcoffee.com (Cloudflare D1 SQL backend) | adzio.io.

Computer Science Tutor (Code Samurai)

Jan. 2026 – Present

Kean University

Union, NJ

- Selected by CS faculty to tutor Database Management Systems in twice-weekly one-on-one sessions for 50+ undergraduates; coach students through **relational database design** and writing efficient **SQL** queries.

Data Visualization & Analytics Intern

July 2026

Group Summer Scholars Research Program, Kean University

Union, NJ

- Lectured daily and mentored 5 high school students under Dr. Ching-yu Huang, teaching relational querying and interactive charting with SQL, MySQL, PHP, JavaScript, and Google Charts.
- Authored the curriculum, lesson slides, and troubleshooting guides that took every student from raw datasets to a published database-driven website and a research poster presented under my guidance, translating analytical concepts for a non-technical audience.